ANNAMALAI UNIVERSITY

FACULTY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

FIRST MID-TERM TEST – FEB-2026

Fourth Semester

(Marks: 30 Marks)

CSBS401- DISCRETE MATHEMATICS

PART A - (4 X 1 =4) Answer ALL Questions

		Marks	CO	PO	BTL
1.	What is mean by tautology?	1	CO1	2,3	K1
2.	Define Rule P and Rule CP in the inference theory	1	CO1	1,2	K1
3.	Show that $\neg P \Rightarrow (P \rightarrow Q)$	1	CO1	1,2	K1
4.	Define the conjunction of two logical statements P and Q	1	CO1	3,4	K1

PART- B (2 x 3 =6 Marks)
Answer either (a) or (b) in each Question

5.	(a) Prove that $P \rightarrow (Q \rightarrow R) \Leftrightarrow (P \wedge Q) \rightarrow R$	3	CO1	1,2	K2
	(OR)				
	(b) Prove that $\neg Q \wedge (P \rightarrow Q) \Rightarrow \neg P$	3	CO1	2,3	K2
6.	(a) Find whether the conclusion C: $\neg P$ follows from the premises $H_1: \neg(P \wedge \neg Q)$; $H_2: \neg R$; $H_3: Q \rightarrow R$ using truth table technique	3	CO1	1,2	K2
	(OR)				
	(b) Prove that $(P \rightarrow Q) \wedge (Q \rightarrow R) \Rightarrow (P \rightarrow R)$	3	CO1	3,4	K2

PART- C (2 x10 =20 Marks)
Answer either (a) or (b) in each Question

7.	(a) Show that $R \rightarrow S$ can be derived from the premises $P \rightarrow (Q \rightarrow S)$, $\neg R \vee P$ and Q	10	CO1	2,3	K3
	(OR)				
	(b) Obtain the PDNF and PCNF of $(\neg P \rightarrow \neg R) \wedge (P \leftrightarrow Q)$	10	CO1	3,4	K3
8.	(a) Show that $S \vee R$ is tautologically implied by $(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S)$	10	CO1	2,3	K3
	(OR)				
	(b) Find whether the conclusion C: R follows from the premises $H_1: P \vee Q$; $H_2: \neg R \rightarrow \neg P$; $H_3: \neg(Q \wedge \neg R)$ using truth table technique	10	CO1	1,3	K3

$\frac{2213}{42} + \frac{2214}{20}$ $\frac{2214}{20} + \frac{2215}{41}$

ANNAMALAI UNIVERSITY

FACULTY OF ENGINEERING AND TECHNOLOGY

ENGINEERING MATHEMATICS

B.E.(CSE) - FOURTH SEMESTER

II- Mid Term Test- April-2026

COURSE: **CSBS401 - DISCRETE MATHEMATICS**

Time: 90 Minutes **Max: 45**

Marks PART- A (6x 1 =6 Marks)

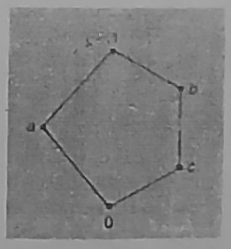
Answer ALL Questions

		Marks	CO	PO	BTL
1.	*If $A = \{1\}$ and $B = \{1,2\}$ find $A \times B$.*	1	CO2	2,3	K1
2.	*Define equivalence relation with an example.*	1	CO2	3,4	K1
3.	*State the modular inequality of a Lattice.*	1	CO3	3,5	K1
4.	*Define Boolean algebra.*	1	CO3	1,2	K1
5.	*Define a Monoid with an example.*	1	CO4	3,4	K1
6.	*What is the identity element of the algebraic system $\{s,*\}$ Where S is the set of integers and * is defined by $a * b = a + b + 2$ for all $a, b \in S$.*	1	CO4	1,2	K1

PART- B (3x3=9 Marks)

Answer either (a) or (b) in each Question

7.	*a) If A, B and C are any three sets prove that $(A \cap B) - C = (A - C) \cap (B - C)$.*	3	CO2	1,2	k2
	OR				
	b) Find all the partitions of the set $A = \{a, b, c, d\}$.	3	CO2	2,3	k2
8.	*a) Draw the Hasse diagram for the divisibility relation on the set D_{24}.*	3	CO3	1,2	k2
	OR				
	b) Verify whether the lattice given by the Hasse diagram is distributive.	3	CO3	2,3	K2



9. a) Show that in a group G,

$(a * b)^{-1} = b^{-1} * a^{-1}$, for all $a, b \in G$

3	CO4	1,2	K3

OR

b) Show that $\{(1,2,3,4,5,6), \times_7\}$ is a cyclic group.

3	CO4	3,4	K3

PART- C (3 x 10=30 Marks)
Answer either (a) or (b) in each Question

10 a) Let $X = \{1,2,3,4,5,6,7\}$ and $R = \{(x,y) | x - y \text{ is divisible by } 3\}$ be a relation defined on X. Show that R is an equivalence relation and draw the graph of R.

10	CO2	1,2	K4

OR

b) Show that for any three sets

$A \times (B \cup C) = (A \times B) \cup (A \times C)$.

10	CO2	3,4	K4

11 a) Show that in a distributive lattice $\{L, \vee, \wedge\}$ if an element $a \in L$ has a complement, then it is unique

10	CO4	2,3	K4

OR

b) Simplify the Boolean function

$f(a,b,c,d) = \sum(0,1,2,3,4,6,7,8,9,11,15)$

10	CO4	2,3	K5

12 a) show that $\{s_3, *\}$, where $s = \{1,2,3\}$ is a group under the operation of right composition of permutation

10	CO5	2,3	K4

OR

b) List the code words generated by the encoding function $e: B^3 \to B^6$ whose parity check matrix is $H = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

10	CO5	2,3	K5

Register Number:

Name of the Candidate:

0229

B.E. DEGREE EXAMINATION, 2025

(COMMON to CS, IT and AI& ML)

(FOURTH SEMESTER)

22CSBS-401/ 22ITBS-401/22AIBS-401: DISCRETE MATHEMATICS

(W.E.F. May 2024)

May] [Time : 3 Hours

Maximum : 75 Marks

PART - A (10x 1 = 10 Marks)
Answer ALL Questions

		Marks	CO	PO	BTL
1.	Define Tautological Implication	1	CO1	1,2	K1
2.	Define the conjunction of two logical statements P and Q	1	CO1	2,3	K2
3.	Define universal relation with example	1	CO2	1,2	K1
4.	Define equivalence relation with an example	1	CO2	1,2	K1
5.	Define a complemented Lattice	1	CO3	2,3	K1
6.	State absorption laws of Boolean algebra	1	CO3	1,2	K1
7.	Define a semi group with an example	1	CO4	1,2	K1
8.	what is the identity element of the algebraic system $\{Q^+,*\}$ Where Q^+ is the set of positive rational numbers and * is defined by $a * b = \frac{ab}{2}$ for all $a, b \in Q^+$	1	CO4	3,4	K2
9.	Define complete graph with an example	1	CO5	1,2	K1
10.	Define spanning subgraph.	1	CO5	1,2	K1

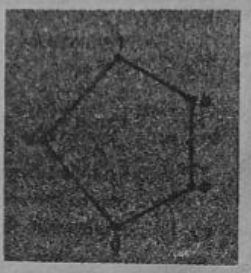
PART- B (5 x 3 = 15 Marks)

Answer either (a) or (b) in each Question

			Marks	CO	PO	BTL
11.	(a)	Prove that $(P \rightarrow Q) \wedge (Q \rightarrow R) \Rightarrow (P \rightarrow R)$	3	CO1	1,2	K2
		(OR)				
	(b)	Obtain the PDNF and PCNF of $(\neg P \vee \neg Q) \rightarrow (P \leftrightarrows \neg Q)$	3	CO1	2,3	K2
12.	(a)	If R and S are equivalence relations on A, show that $R \cap S$ is also an equivalence relation	3	CO2	2,3	K2
		(OR)				
	(b)	If $A = \{1,2,3\}$ and $R = \{(1,1),(1,2),(2,1),(2,3),(3,1),(3,2)\}$ $S = \{(1,1),$	3	CO2	2,3	K2

$\{(1,2),(2,1),(2,2),(3,2),(3,3)\}$,
be a relations defined on A. Find $R \circ S$, $R \oplus S$ and R^2

13. (a) Verify whether the lattice given by the Hasse diagram is distributive | 3 | CO3 | 1,2 | K2 |



(OR)

(b) Simplify the Boolean expression $xy' + z + (x' + y)z'$ | 3 | CO3 | 2,3 | K2 |

14. (a) Prove that $\{Z_6, \times_6\}$ is a monoid | 3 | CO4 | 1,2 | K2 |

(OR)

(b) Show that $\{(1, -1, i, -i), \times\}$ is a cyclic group. What are its generators? | 3 | CO4 | 3,4 | K2 |

15. (a) Define Eulerian and Hamiltonian graphs with example | 3 | CO5 | 1,2 | K2 |

(OR)

(b) Draw the graph whose adjacency matrix is $\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$ | 3 | CO5 | 2,3 | K2 |

PART- C (4 x10 = 40 Marks)

Answer either (a) or (b) in each Question

16. (a) Find whether the conclusion C: R follows from the premises $H_1: P \vee Q$; $H_2: \neg R \rightarrow \neg P$; $H_3: \neg(Q \wedge \neg R)$ using truth table technique | 10 | CO1 | 2,3 | K3 |

(OR)

(b) Show that $t \wedge s$ can be derived from the premises $(p \rightarrow q)$, $q \rightarrow \neg r$, r, $p \vee (t \wedge s)$ | 10 | CO1 | 3,4 | K3 |

17. (a) Show that for any three sets $A \times (B \cup C) = (A \times B) \cup (A \times C)$ | 10 | CO2 | 2,3 | K3 |

(OR)

(b) If R is the relation on the set of ordered pairs of positive integers such that $(a, b) \in R$ if and only if $3a + 4b = 7n$ for some integer n Show that R is an equivalence relation | 10 | CO2 | 1,3 | K3 |

18. (a) Show that $(a + b')(b + c')(c + a') = (a' + b)(b' + c)(c' + a)$ in a Boolean algebra. | 10 | CO3 | 1,2 | K3 |

(OR)

(b) Simplify the Boolean function
$f(a, b, c, d) = \Sigma(0,1,2,3,4,5,6,7,8,9,11)$ | 10 | CO3 | 2,3 | K3 |

19. (a) Prove that the set {1,3,7,9} is an abelian group under multiplication modulo 10 | 10 | CO4 | 2,3 | K3 |

(OR)

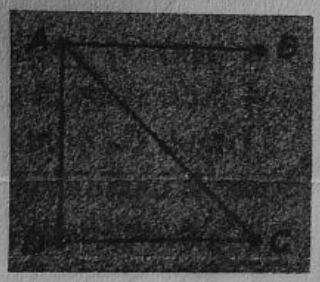
(b) List the code words generated by the encoding function | 10 | CO4 | 2,3 | K3 |

$e: B^2 \rightarrow B^5$ whose parity check matrix is $H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

PART- D (1 x 10 = 10 Marks)

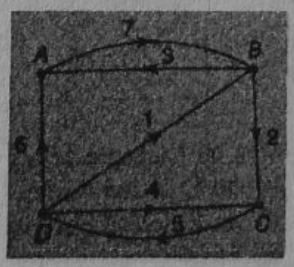
Answer either (a) or (b)

20. (a) Summarize all the paths of length 4 between A and C for the following graph using adjacency matrix | 10 | CO5 | 2,3 | K4 |



(OR)

(b) Evaluate the shortest distance matrix and the corresponding shortest path matrix for all the pairs of vertices in the weighted graph given below using Warshall's algorithm. | 10 | CO5 | 2,3 | K4 |



A-B-A-D-C

A-B-A-B-C

A-D-A-D-C

A-D-A-D-C

A-C-A-D-C

A D-C-D-C

A D-C-A-C

A-D-C-A-C